~~Sonderfälle / Rand~~

UnGültige Bereiche:

(1) Phi == 0 . Keine Strahlung. T = 0 ?

(2) $\frac{1}{\Phi} + 1 <= 0$ (logarithmus von negativer Zahl)

$\Longleftrightarrow$

$\Phi \leq -1$: Strahlung kleiner/gleich -1. T = ?

Equations used in Civmain / 66R-Daq:

$x :=$ Digital value

$x_0 :=$ Digital offset

constants $A, B, C, x_0, \Delta\phi, \varepsilon, \tau$

$$\phi(x) = \frac{x - x_0}{A \cdot \varepsilon \cdot \tau} - \Delta\phi \qquad (1)$$

$$T(x) = \frac{B}{\ln\left(\frac{\frac{1}{\phi(x)} + 1}{C}\right)} \qquad (2)$$

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(2) solved for $\phi(x)$

$$\phi(x) = \frac{1}{C \cdot e^{\frac{B}{T}} - 1}$$

Plancks law. Spectral radiance B_ν

$\lambda :=$ Unit wavelength

$$B_\nu(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{e^{\left(\frac{h \cdot c}{\lambda \cdot k_B} \cdot \frac{1}{T}\right)} - 1}$$